



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**
WEEKLY PROGRESS REPORT (10/08/2026 – 16/08/2026)

WEEK 2

PROJECT NAME

**CHATBOT FOR STRUCTURED, UNSTRUCTURED
AND SEMANTIC KNOWLEDGE QUERY**

PROJECT DESCRIPTION :

**MAKING A CHATBOT CAN GIVE ANSWER TO
QERY FOR DIFFERENT TYPES OF DATA**

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COLLEGE NAME :

BITS PILANI

10/08/2026	• Tried different AI models for our project. • Study different models for different data queries like structured, unstructured, knowledge graphs.
11/08/2026	• Learn about LLM orchestration and frameworks like Langchain, LanghGraph for implementation.
12/08/2026	• Tried to learn Langchain by making a simple chatbot that query things with the help of prompt ingestion.
13/08/2026	• Studied all types of data structures like structured, unstructured and knowledge graphs and tried to prepare dataset for model training.
14/08/2026	• Try making a simple model that can query from different datasets by making decisions on its own.
15/08/2026	Holiday (INDEPENDENCE DAY)
16/08/2026	Holiday (Sunday)

Next Week Plan	I am planning to explore more RAG and LLM orchestration and will try to make simple prototype.
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REFERENCE:

- <https://docs.langchain.com/>
- <https://docs.langchain.com/oss/python/langgraph/overview>
- <https://docs.python.org/3/library/frameworks.html>
- <https://docs.pytorch.org/docs/2.13/index.html>
- v

SCREEN SHOTS:

```

1  # EV Powertrain Fault Intelligence System
31 ## Quick Start (100% Free / Local)
45 ### 2. Start all services
47 docker compose up -d
48
49 This starts PostgreSQL + pgvector, Neo4j, Ollama, and the FastAPI app.
50 On first run, ollama will pull `llama3.2` and `nomic-embed-text` (~5 GB).
51
52 ### 3. Verify everything is running
53 ```bash
54 curl http://localhost:8058/api/v1/health
55 # → {"status":"ok","postgres":true,"neo4j":true}
56 ```
57
58 ### 4. Ingest a document
59 ```bash
60 docker compose exec app python -m app.ingestion.ingest \
61 --file data/docs/your_manual.pdf \
62 --title "EV Battery Manual" \
63 --vehicle "Tesla Model 3"
64 ```
65
66 ### 5. Ask a diagnostic question
67 ```bash

```

Microsoft Windows [Version 10.0.26200.8737]
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D:\projects\ev-agentic-ai>curl http://localhost:8058/api/v1/health
curl: (7) Failed to connect to localhost port 8058 after 2235 ms: Could not connect to server

D:\projects\ev-agentic-ai>
* History restored

Microsoft Windows [Version 10.0.26200.9168]
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```

async def lifespan(app: FastAPI) -> AsyncGenerator[None, None]:
    logger.warning("Neo4j unavailable (graph features disabled): {exc}")

    logger.info("System ready.")
    yield

    logger.info("Shutting down ...")
    await DatabasePool.close()
    await GraphiticClient.close()
    logger.info("Shutdown complete.")

app = FastAPI(
    title="EV Powertrain Fault Intelligence System",
    description="Agentic RAG + Knowledge Graph + Time-Series for EV diagnostics.",
    version="1.0.0",
    lifespan=lifespan,
    docs_url="/docs",
    redoc_url="/redoc",
)

app.add_middleware(

```